Technical Report No. 27

Pharmaceutical Package Integrity

PDA Journal of Pharmaceutical Science and Technology

1998

Supplement

Volume 52

Number S2

PDA

An International Association for Pharmaceutical Science and Technology

PDA Task Force on Container/Closure Integrity

Michael C. Carroll, The Liposome Company, Inc.
Vivian F. Denny, Peak to Peak Pharmaceutical Associates
Dana M. Guazzo, Ph.D., The R. W. Johnson Pharmaceutical Research Institute
Mark W. Kaiser, Lancaster Laboratories
Lee E. Kirsch, Ph.D., The University of Iowa
John D. Ludwig, Ph.D., G. D. Searle and Company
Gerald K. Masover, Genentech, Inc.
Jeri L. May, DuPont Merck Pharmaceutical Co.
Jeanne E. Moldenhauer, Ph.D., Fujisawa USA, Inc.
Jean I. Olsen, SmithKline Beecham Pharmaceuticals
Terri M. Polson, Glaxo Wellcome Inc.
Glenn E. Wright, Eli Lilly & Co.

Pharmaceutical Package Integrity

Technical Report No. 27

PDA

April 1998

TECHNICAL REPORT NO. 27
PHARMACEUTICAL PACKAGE INTEGRITY

Table of Contents

1. Preface 1

2. Introduction 1

3. Integrity Assurance Through Product Development And Product Life 2
   - 3.1 Package Design, Process Development, and Durability Assessment 2
   - 3.2 In-Process Control 3
   - 3.3 Product Testing 4

4. Package Integrity and Leak Rate Specifications 5

5. Test Method Selection 6
   - 5.1 Test Method Sensitivity 6
   - 5.2 Other Test Method Selection Considerations 7

6. Methods of Establishing Microbial Package Integrity 8
   - 6.1 Microbial Challenge vs. Physical Test Methods 8
   - 6.2 Liquid Immersion Microbial Challenge Tests 9
   - 6.3 Airborne Microbial Challenge Tests 9
   - 6.4 Physical Leak Test Detection Methods 10
     - 6.4.1 Direct Correlation to Microbial Ingress 10
     - 6.4.2 Indirect Correlation to Microbial Ingress 10

7. Decision Tree 11

8. Leak Test Methods 15
   - 8.1 Bubble Test 16
   - 8.2 Helium Mass Spectrometry 17
   - 8.3 High Voltage Leak Detection (HVLD) 20
   - 8.4 Lid Deflection 21
   - 8.5 Liquid Tracer 22
   - 8.6 Microbial Challenge, Aerosolization 25
   - 8.7 Microbial Challenge, Liquid Immersion 26
   - 8.8 Package Headspace Analysis 28
   - 8.9 Particulate Transmission 29
   - 8.10 Residual Gas Ionization 30
   - 8.11 Tracer Gas Leak Detection 31
   - 8.12 Ultrasonic Detection of Gas Flow 33
   - 8.13 Ultrasonic Imaging 34
   - 8.14 Vacuum/Pressure Decay 35
   - 8.15 Vacuum/Pressure Retention 37
   - 8.16 Visual Inspection 38
   - 8.17 Volumetric Gas Flow 39
   - 8.18 Weight Loss/Gain 40

9. Bibliography 42

1. Preface

This report reviews issues of pharmaceutical product package integrity and provides guidance for evaluating the barrier qualities of a pharmaceutical package. It supersedes the previously issued "PDA Technical Information Bulletin No. 4, Aspects of Container/Closure Integrity." In this updated guideline, the term "product package" is used, rather than the term "container/closure system," as it encompasses any nonporous barrier system that protects a pharmaceutical product from the environment. Similarly, the term "package seal" is used generically to mean all types of seals or closure systems. Although it is written to reflect the complexity of all pharmaceutical products and packages, the emphasis throughout the document is clearly on packaging intended for sterile products.

This report was written by the following members of the PDA Task Force on Container/Closure Integrity:

Michael C. Carroll, The Liposome Company, Inc.
Vivian F. Denny, Peak to Peak Pharmaceutical Associates
Dana M. Guazzo, Ph.D., The R. W. Johnson Pharmaceutical Research Institute
Mark W. Kaiser, Lancaster Laboratories
Lee E. Kirsch, Ph.D., The University of Iowa
John D. Ludwig, Ph.D., G. D. Searle and Company
Gerald K. Masover, Genentech, Inc.
Jeri L. May, DuPont Merck Pharmaceutical Co.
Jeanne E. Moldenhauer, Ph.D., Fujisawa USA, Inc.
Jean I. Olsen, SmithKline Beecham Pharmaceuticals
Terri M. Polson, Glaxo Wellcome Inc.
Glenn E. Wright, Eli Lilly & Co.

In addition, the Task Force acknowledges the immense help and support of Michael S. Korczynski, Ph.D. and Russell E. Madsen of PDA. Their editorial assistance and their insight into many of the key issues of package integrity were invaluable contributions to this effort.

This document should be considered as a guide; it is not intended to establish any mandatory or implied standard.

2. Introduction

Product quality can be defined in terms of the ability of a product to perform its desired functions despite the stresses to which it has been exposed before and during its intended use. Pharmaceutical container systems may have a variety of functions including facilitation of product use, prevention of improper use, light protection, containment of the product, maintenance of a microbial barrier, and maintenance of a rarefied product environment. Assurance of package integrity is primarily concerned with product containment and the maintenance of an effective barrier against ingress of microorganisms or potentially reactive gases such as oxygen or water vapor, and sometimes, the maintenance of a vacuum. The information provided in this guideline is intended to assist users in developing integrity assessment strategies for use during the phases of product life.

3. Integrity Assurance Through Product Development And Product Life

Product package integrity assessment begins during a product's development phase and continues throughout its life cycle, encompassing routine manufacturing and shelf-life stability assessments.

The phases of a product's life cycle include:

- 3.1 Package design, process development, and durability assessment
- 3.2 In-process control
- 3.3 Product testing

Further reconfirmation of product package integrity should occur when there are major changes in the package system that may affect its integrity, including package design, package materials, and/or when changes occur in the manufacturing processing conditions, including sterilization.

The following subsections contain information on the critical factors for assessing a product's package integrity over its life cycle.

During the package design/process development phase, a thorough knowledge and understanding of the package seal is critical for designing an appropriate package system. Subsection 3.1 describes the nature of these requirements, including a knowledge of the seal location, the sealing mechanism, and the critical factors of design, assembly, and processing that may impact seal quality. The importance of testing the package and processing parameters at extremes is stressed to ensure that the package will be able to withstand the rigors of processing, distribution, and usage.

Once a product reaches commercial manufacturing, in-process control of critical production and processing steps are necessary. Subsection 3.2 describes these vital checks in greater detail.

Product leak testing is discussed in subsection 3.3. The types of methods available for product testing are included, along with their limitations.

3.1 Package Design, Process Development, and Durability Assessment

Package integrity quality assurance involves the selection of appropriate packaging materials, the design of the [seal or] closure system, and the verification that the incoming components meet specifications. Establishing leakage rate specifications may be desirable early during product development based on the particular product and package performance characteristics to ensure that the product will reflect these limits. Integrity tests used during package design may include gross leak tests that simply check for leak presence or location, but should also consist of those that test at or near the established leakage/permeation rate specification.

Packages may be closed via either seals or closures. By definition, a seal is the result of the bonding of two similar or dissimilar surfaces. Examples of seals include: 1) contiguous containers formed from a single material (for example, glass or plastic ampoules); 2) the direct fusion of two material surfaces, usually by heat (for example, heat sealed plastic bags or unit-

dose blister packaging for solid oral dosage forms); and 3) the bonding of two material surfaces with an intermediate adhesive agent. A closure, on the other hand, is the physical mating of two surfaces. In the strictest sense, all closures are tortuous path barriers. The degree to which a closure can limit leakage is a function of the closure design, the viscoelastic nature of the package components, and the extent of package component fit. For instance, an elastomeric rubber stoppered sterile vial is capable of achieving a very low gaseous leakage rate due to the viscoelastic nature of the rubber and its ability to conform to the vial surface upon compression. Other closure systems, such as metal on metal or plastic on metal, rely on the frictional mating of relatively incompressible materials. Still other closures do not depend on component fit as much as the existence of a tortuous path. In this guideline, the term "package seal" will be used generically to mean all seal and closure system types.

There are critical factors that impact the quality of each package seal. Such factors typically include package component variables of composition, dimension, and critical defects; and sealing operation variables of time, temperature, pressure, and torque. The impact of processing on package integrity must also be characterized. Package component washing, drying, siliconization and sterilization, and product processing such as terminal sterilization may all impact the ability of the package seal to maintain its effectiveness.

During the package design and process development phases, the acceptable ranges for each critical variable should be established based on package performance criteria that incorporate integrity testing. Wherever feasible, the variables of design and processing should be exaggerated and their impact on package integrity evaluated to help define the margin of safety for that variable. Such factors may include, for example, time, temperature, and pressure variables for heat seal operations and for steam sterilization cycles. Stability studies extending through shelf-life must also include integrity tests of the packaged product produced under typical manufacturing conditions. In some cases it may also be prudent, although not mandatory, to place product on stability that has been manufactured at critical processing or package design extremes. When greater time and effort are spent on package integrity verification during the design and process development phases, fewer problems will likely be encountered after commercialization.

The assurance of product quality depends on the ability of the product to maintain integrity throughout the stresses anticipated during storage, distribution, and use. During product development, the effects of various environmental and handling stresses may need to be considered in addressing package integrity. Relevant types of stress may include temperature, pressure, and relative humidity extremes, shock and vibrational stress, and exposure to microorganisms. The evaluation of package integrity may include exposure of the product to simulated conditions of storage, distribution, and use coupled with appropriate leakage detection methodology. The integrity tests chosen are generally those that detect sporadic damage to the product, which may result in gross leakage or product loss, and in the case of sterile products, microbial ingress.

3.2 In-Process Control

The processes that have been determined during product development to most critically impact package integrity should be those that are targeted for in-process control. Performance criteria

for in-process controls are typically based on the performance of a specific unit operation rather than the performance of the final product. For example, in-process controls and tests may be used to statistically verify that the time, temperature, and pressure variables of a sealing operation are "in control" as opposed to final product testing used to demonstrate that microbial barrier or rarefied internal atmosphere criteria have been met. During validation studies it must be demonstrated that a process "in control" yields only products that meet the established leakage rate specifications.

For a limited number of products, instrumentation exists that permits high-speed 100% on-line verification of package integrity at the established leakage rate specifications. But for many sterile products in-process leak test instruments have limitations that preclude high speed, on-line integrity verification at the required product leakage rate specifications. In-process leak test instruments are designed to check for intermittent defective packages leaking at a gross rate that circumvented the various in-process controls. Certainly no instrumentation currently available for in-process use is intended to measure permeation rates of gases through the package itself. In-process testing is strictly limited to leakage through package imperfections.

There are a variety of logical, validatable in-process integrity tests that can help verify product/package integrity. However, no amount of in-process or final product testing can replace the need for the control and monitoring of various critical package and sealing parameters. For example, vision systems can be used to cull out defective components, gas flow rate meters can be installed on ample sealing lines, time/temperature/pressure controls can be used on thermal sealing equipment, and torque monitors can be placed on screw-capping equipment. These on-line in-process controls, in combination with well-designed product/process validation studies, are an essential part of the total package integrity quality assurance.

3.3 Product Testing

A product that has been properly and carefully designed with critical production processes well characterized and controlled should have little need for final product release testing for package integrity. Certainly the quality of a pharmaceutical product cannot merely be based on the results of a product integrity test, especially when such tests are usually performed on a fraction of the production lot. Sterility tests have historically been considered an indicator of package integrity; however, due to the insensitive nature of sterility tests, the small number of samples tested, and the possibility of false positives, these results are no longer considered reliable as a final indicator of package integrity.

Generally, integrity test methods selected for final product testing are designed to verify the absence of gross leaks well above permeation rate levels. For those package systems that can be tested 100% on-line with instruments shown to measure near or below the leakage rate specification there is no need for additional product integrity testing. For other products a statistically established percentage of the batch may be integrity tested for in-house process control purposes.

Indirect tests of package integrity may also be employed on production lot samples. For example, indirect measures of integrity may include visual inspection, residual seal force of a serum vial with elastomeric closure, or seal strength of an adhesive or thermal seal. The

acceptance criteria for such tests should be based on the critical variable acceptance ranges established during the development phases of package design and process development.

4. Package Integrity and Leak Rate Specifications

This section provides a definition of leaks and leakage, and presents examples of leakage rate units of measure. The concept that all packages leak to some extent is introduced, although such leakage may not have a negative impact on product quality. Therefore, when designing a product/package system, a leak rate specification must be established. Examples of leak rate specifications are provided.

A leak is an unintended crack, hole, or porosity in a package capable of passing a liquid or gas through the package. Leakage is the movement of the liquid or gas through the leak. Leakage is defined in units of mass or volume flow of a gas or liquid over time under defined conditions of temperature and pressure. Leakage is often described in units of standard cubic centimeters of gas (air) that can move through an orifice per second (std cc/sec or sccs[1]) under defined conditions of temperature and differential pressure. Similarly, leakage units of atm cc/s are also used to define leakage measured at local atmospheric pressure and temperature. While the two are not identical, units of sccs and atm ccs are often used interchangeably. Another common unit of measure is the SI unit of Pa m^3/s, or Pascal cubic meters per second. One Pa m^3/s is equal to 9.87 sccs or roughly 10 sccs. Various general references on the subjects of leakage, leak test methods, and units of measure are referenced at the end of this document.

Leakage rates from a pharmaceutical package may vary over a wide range. For instance, leakage through an orifice large enough to create an audible sound may be approximately 1 sccs. The human eye can detect visible flow of liquid occurring through holes that leak at gaseous rates of 0.1 sccs. No visible water will leak when dry air, at the same pressure, leaks at a rate as great as 10^{-4} sccs. Leak detection by helium spectroscopy is the most sensitive leak testing method currently available and, under certain conditions, can detect leakage as small as 10^{-11} sccs (Ref. ASNT, 1982, and Varian, 1995).

Leakage need not always be described in units of gas mass or volume flow over time. It may also be described in other terms believed to be more meaningful to the specific product or leak test method. For instance, leakage rate may be described in units of product weight loss or gain, or rise of percent oxygen in an inert gas-flushed headspace, or as the volume of dye that has migrated through a package seal. Regardless of the leakage units used, it is important to clearly indicate the relevant leak test parameter conditions. For instance, weight loss might be described as grams of product loss measured over 2 years when the product is stored at 35°C. Such descriptions of leakage that clearly and meaningfully define mass or volume flow over time are as acceptable as gas leak rate units.

---

[1] 1 std cc = 1 standard unit volume of gas (air) at sea level atmospheric pressure (101.3 kPa, 760 mm Hg or 760 torr), and 20°C (293°K) temperature.

Strictly speaking, all packages demonstrate leakage to some extent, although such leakage may not affect product quality. Therefore it is important to establish an allowable leakage rate for a given product/package system early during product development. The product's needs for protection will dictate the performance criteria of a package and therefore the allowable leakage rate. All pharmaceutical packages must contain their intended product, and thus must prevent gross leakage. For a nonporous package this may equate to a maximum gaseous leakage rate of approximately 10^{-3} sccs (Ref. ASNT, page 699). Packages designed to only prevent aerosolized microbial challenge may be able to tolerate a higher leakage rate specification.

Sterile liquid products require containers that prevent minute liquid product leakage and microbial ingress. Other products, such as sterile powders, ointments, and lyophilized products, may also require containers that prevent liquid-borne microbial ingress. For all of these products, the leakage rate specification limit should be the same as the cut-off for liquid passage, as microorganisms are believed incapable of penetrating a leak smaller in size than can be penetrated by liquids (Ref. Kirsch and Morton). Water is generally thought to be incapable of penetrating pores that allow less than approximately 10^{-5} sccs of airflow at a differential pressure across the seal of approximately 1 atmosphere. The relationship between gaseous flow, liquid leakage, and liquid-borne microbial ingress was demonstrated by L. Kirsch, et al and Morton, 1989. For other products, much smaller rates of leakage or even permeation through the package may cause an unacceptable loss of a critical gas headspace or vacuum, or an excessive migration of water vapor or solvents.

Once the level of acceptable leakage/permeation has been established, a leakage/permeation specification incorporating a reasonable margin of safety may be set.

5. Test Method Selection

This section is devoted to leak test method selection criteria. In subsection 5.1 the need to select tests of adequate sensitivity is discussed. Because test methods have an optimum sensitivity range, it may be necessary to select more than one type of leak method for a single product/package system. Subsection 5.2 discusses other test method selection considerations.

5.1 Test Method Sensitivity

In selecting a leak test method it is important to choose a test which is capable of detecting the leakage rates of concern. The leakage rate of concern will depend on the product/package system as well as the phase of the product life cycle in which the test will be utilized. During certain times in the product life cycle, a leak test method is needed merely for gross leakage detection or for leakage location. But at some point in the development of a product-package system, the leak test method selected must be capable of detecting leakage at the product's established leakage rate specification.

Most physical leak test methods have an optimum sensitivity range, meaning that a method capable of detecting leaks between 10^{-6} sccs and 10^{-2} sccs may not be appropriate for gross leaks of 10^{0} sccs. Some physical test methods have a "gross" and a "fine" leak test mode in order to expand this sensitivity window.

However, it is inappropriate and can be quite costly to select a test method that is too sensitive. For instance, a helium leak test capable of detecting leaks as low as 10^{-11} sccs would be an inappropriate test for a high density polyethylene screw-capped bottle merely intended to retain tablets and capsules. But the converse is also true; an airborne microbial challenge test would be meaningless for a product with a gaseous leakage rate specification of 10^{-7} sccs.

Because leak test methods have a range of sensitivity, and because the leakage rate of concern may differ depending on the stage of the product's life cycle, it may be necessary to test a given product using more than one leak test method, each designed to detect leaks of varying magnitude. A package containing a lyophilized powder undergoing shipping simulation studies may need to be tested for larger leaks that have resulted from shock or vibration damage. While long term stability samples of the same product may need to be checked for leaks indicative of gaseous leakage and permeation. Once in production, a gross leak test detection system may be most appropriate for a high speed filling line where package breakage is a concern.

A test method's sensitivity can be expressed in various units of measure. Optimum sensitivity of a method can only be realized through careful selection of the test parameter conditions. If, for example, an oxygen analyzer is used to monitor the headspace of a package, the instrument's sensitivity is expressed as the smallest change in headspace oxygen content detectable by the gas analyzer. But the ability of this method to detect very small leakage/permeation rates will depend on the time that has elapsed since the package was first sealed, and on the extent of package exposure to an oxygen partial pressure differential. As another example, the sensitivity of a dye test is described as the smallest volume of dye that can be detected in the given product-package under the stated test conditions of time, temperature, and pressure. Dye tests can be optimized by maximizing the test conditions, as well as by manipulating the dye formulation and detection technique. Lastly, the sensitivity of a helium mass spectrometry leak detection method is described as the smallest measurable leak rate in units of standard cc of gas (helium) per second. Helium leak detection capabilities can be improved by maximizing the concentration of helium in the package headspace, and by choosing the most optimum detection mode.

5.2 Other Test Method Selection Considerations

A wide variety of leak test methods are available for the evaluation of pharmaceutical packages. In Section 8, "Leak Test Methods," summaries of several of the most commonly used physical methods are provided. The usefulness of a physical method must be determined on a product-by-product basis by evaluating the ability of a method to test the seal of interest, the sensitivity of the instrument or method, and any relevant sources of measurement variation. In some cases factors such as data storage, analysis and display functions, statistical manipulation of data, practicality, speed, safety, on-line integration feasibility, and expense may play important roles in the selection of the most appropriate leak test method.

Sterile package integrity has traditionally been verified by airborne or liquid immersion microbial challenge tests. Microbial challenge tests have a long history of use and may be an indicator of microbial package integrity. Also in Section 8, a brief description of microbial challenge tests is provided. Section 6, "Methods of Establishing Microbial Package Integrity,"

reviews both the microbial challenge approach and the alternate physical testing approach for verifying a package's microbial barrier properties.

6. Methods of Establishing Microbial Package Integrity

All sterile pharmaceutical packages must prevent the ingress of microorganisms. This section focuses on liquid immersion and airborne microbial challenge test methods and alternate physical test methods useful for establishing a package's microbial barrier properties. In subsection 6.1 the rationale for selecting a microbial challenge test versus a physical test method is discussed. Subsections 6.2 and 6.3 describe how one may directly verify a package's microbial integrity through either liquid immersion or airborne microbial challenge tests.

In subsection 6.4 an outline is presented describing how physical leak tests may be validated as an alternative indicator of package microbial integrity. A direct correlation of physical integrity test results to microbial barrier properties can be done by exposing a given population of packages to both physical and microbial challenge tests (subsection 6.4.1). An indirect approach is to compare the physical method's sensitivity to leakage rates previously demonstrated as preventing microbial ingress (subsection 6.4.2).

6.1 Microbial Challenge vs. Physical Test Methods

For most sterile pharmaceutical packages, the leakage rate specification is that which will prevent the ingress of microorganisms. Such packages must therefore be validated as providing an effective barrier against either airborne and/or liquid-borne microbial ingress. The microbial barrier properties of a package may be demonstrated using either microbial challenge tests or physical tests that have been correlated to microbial ingress.

There are several reasons for choosing to use a microbial challenge test. First, there may be no reasonable alternative to microbial challenge tests. For instance, certain closures or packages that rely solely upon a tortuous path to prevent airborne microbial ingress may not be reliably tested by a physical method. A second reason may be that the package design does not lend itself to sufficiently sensitive physical test methods. For example, a package may not tolerate the relatively high differential pressures required for testing by vacuum decay or tracer gas techniques. A third reason may be that the manufacturer does not have the resources or expertise required for developing and validating an alternate physical test method. It is also possible that the product is only intended for limited distribution so that an extensive physical test method validation program cannot be justified.

Conversely, a physical test may be the preferred approach for microbial barrier package validation. If a sufficiently sensitive physical test method is available that is compatible with the product-package system, it is often more economical in the long run to invest in developing and validating such a physical test. Physical tests can be selected that yield qualitative or quantitative results or that can pinpoint the location of leaks. Microbial challenge tests are relatively insensitive, yield inconsistent results, are labor intensive, necessitate extensive storage space for sample incubation, and require long test times. Another disadvantage is that microbial

challenge tests must often be done using media-filled packages rather than packages containing the intended product.

Physical tests are the more logical choice when the leakage rate specification for a product/package system is significantly lower than the limit for microbial ingress. For example, a given sterile glass vial closed with an elastomeric stopper may require a leakage rate specification of 10^{-7} std cc/s to maintain an inert gas headspace. The gaseous leakage rate that correlates to the ingress of liquid-borne microorganisms for this package system has been shown to be approximately 10^{-5} sccs when tested by helium mass spectrometry (Ref. Kirsch, et al). Given this scenario, it is unnecessary to further establish the package's microbial integrity by microbial challenge tests. In fact, gaseous leakage below the rates that allow microbial ingress can be assured for some packages by verifying the maintenance of an inert gas headspace when the packages are tested over an extended period of time, and when they are sufficiently challenged with a differential pressure leakage driving force.

6.2 Liquid Immersion Microbial Challenge Tests

There is no standard liquid immersion microbial challenge test, but the following guidelines are suggested for the establishment of a meaningful and useful method. First, it is important to select a small bacterium that is demonstrated to yield a high population of viable organisms at the end of the study. Whenever possible, positive controls that demonstrate growth must be included. These may consist of defective packages or packages prepared outside the package assembly process control limits. As with any package integrity study, a statistically representative number of samples should be tested. When packages vary in size, but not in design or package component materials of construction, worst case container sizes are to be evaluated, or, preferably, sizes should be bracketed. If more than one seal is relied upon for package integrity, each seal should be challenged. The position of the container during the integrity test should ensure that the seal is exposed to the challenge microorganisms. The media, incubation, and challenge conditions of time and pressure must be logically chosen keeping in mind critical integrity variables such as package component materials, assembly, processing parameters, and challenge organism viability and morphology. Whenever reasonable, the product itself should be challenged, as it is possible the product may influence seal integrity. However, this may not be feasible based on the product's inability to support microbial growth.

6.3 Airborne Microbial Challenge Tests

There are many references describing airborne microbial challenge test procedures published by the food and medical device industries. Several of these are listed in the references at the end of the Microbial Challenge Aerosolization Test Method summary. It is recommended that such references be utilized in establishing an airborne microbial challenge test for pharmaceutical packages. Very briefly, some critical factors in the design of such tests include location and position of the microbial challenge source versus the test articles, methodology used to aerosolize the microbial challenge, size and morphology of the challenge organisms, and the size, shape, and configuration of the test chamber.

There are many variables that affect the outcome of airborne microbial challenge tests. The key to performing consistent airborne challenges is design of the aerosolization apparatus with

sufficient fine adjustments and system monitors (for example, pressure gauges, flow meters) so that variation between runs and/or operators is minimized.

6.4 Physical Leak Test Detection Methods

Unlike microbial challenge tests, physical leak test methods do not directly measure microbial ingress, but rather, they measure some physical property of the leak that may be related to microbial failure. The relationship between the sensitivity of a physical leak test method to the probability of microbial failure can be either directly or indirectly determined.

6.4.1 Direct Correlation to Microbial Ingress

Typically the development of a direct correlation between a physical integrity test and microbial ingress first requires the creation of representative acceptable and defective packages that can be tested by both leak test methods. Then it must be experimentally demonstrated that the measured leak rate by the physical method can be quantitatively correlated to the probability of microbial failure.

To illustrate, the following results would need to be demonstrated by direct correlation of a liquid chemical tracer test to a microbial immersion test:

1. the presence of tracer in all test units that show microbial ingress,
2. the presence of tracer in some test units that fail to show microbial ingress,
3. the absence of tracer in some test units that fail to show microbial ingress.
4. the absence of microbial ingress in all of the test units that do not show tracer.

A direct correlation of tracer to microbial ingress would need to take into consideration the potential interference of the tracer with microbial growth, and the potential interference of microbial growth on tracer detection.

Liquid chemical tracer tests provide simple pass-fail results; either the tracer is present or it is not. If a physical test method is used that provides a range of leakage rate results, correlation to microbial ingress is based on the rate of leakage determined by the physical method versus the probability of microbial failure.

6.4.2 Indirect Correlation to Microbial Ingress

A physical test method can be indirectly correlated to microbial ingress based on documented evidence of microbial ingress within the physical method's leakage rate detection range. Again, for a sterile glass vial with elastomeric stopper, it has been demonstrated that the helium leak rate (under specific testing conditions) that corresponds to microbial ingress as determined by rigorous immersion challenge testing is 10^{-5} sccs (Ref. Kirsch, et al). This information can be used to verify the microbial ingress leak detection capability of another physical method by demonstrating that the second method can reliably detect leakage at or below the documented leakage rate of about 10^{-5} sccs. It is important to keep in mind that the leak test gas and the test conditions of temperature and differential pressure used for the referee test (in this case the helium test) may differ significantly from those employed by the alternative physical method. Therefore it may be necessary to calculate an equivalent leakage rate (either > or < 10^{-5} std cc/s) against which the alternative physical leak test method must be compared.

Indirect correlation of a physical test method to liquid-borne microbial challenge tests can also be made by demonstrating that the physical method is capable of detecting microscopic liquid passageways near the limit of microbial ingress. This relies on the assumption that without a passageway large enough to permit liquid flow, microbial ingress cannot occur. For example, a dye or liquid chemical tracer can be used as an indirect indication of microbial package integrity. In order to rely on such results, the sensitivity of the dye or liquid chemical tracer test must be optimized in order to approach the leakage limit for liquid ingress.

7. Decision Tree

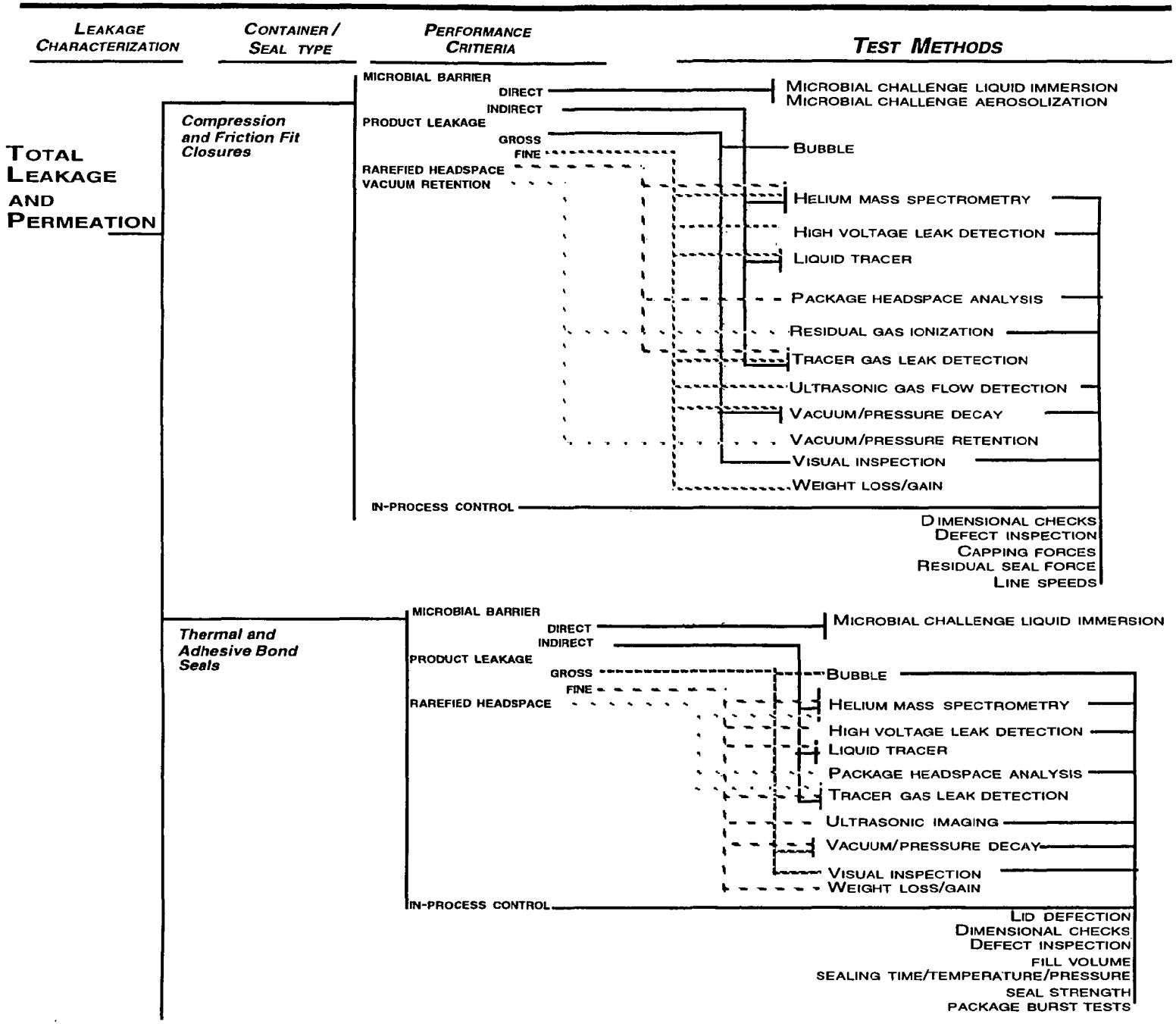
A decision tree is presented in this section to assist in selecting appropriate leak tests and process control methods. When using the tree, it must first be decided whether leakage/permeation is to be measured, or whether the leak location is to be pinpointed. Within the category of "Total Leakage and Permeation" measurement, there is a further classification according to seal type. These include compression & friction fit closures, thermal & adhesive seals, contiguous containers, or tortuous path closures. (These seal types are defined more fully at the end of this Section.) Each seal type can be intended to effect any one of a variety of package integrity performance criteria. The seal or closure may be intended as a microbial barrier, or it may be designed to restrict product leakage, or to maintain a rarefied gas or vacuum headspace. Another subheading "In-process Control" is also included under "Performance Criteria" to help identify some of the process monitors and controls used to ensure package quality and minimize package integrity problems.

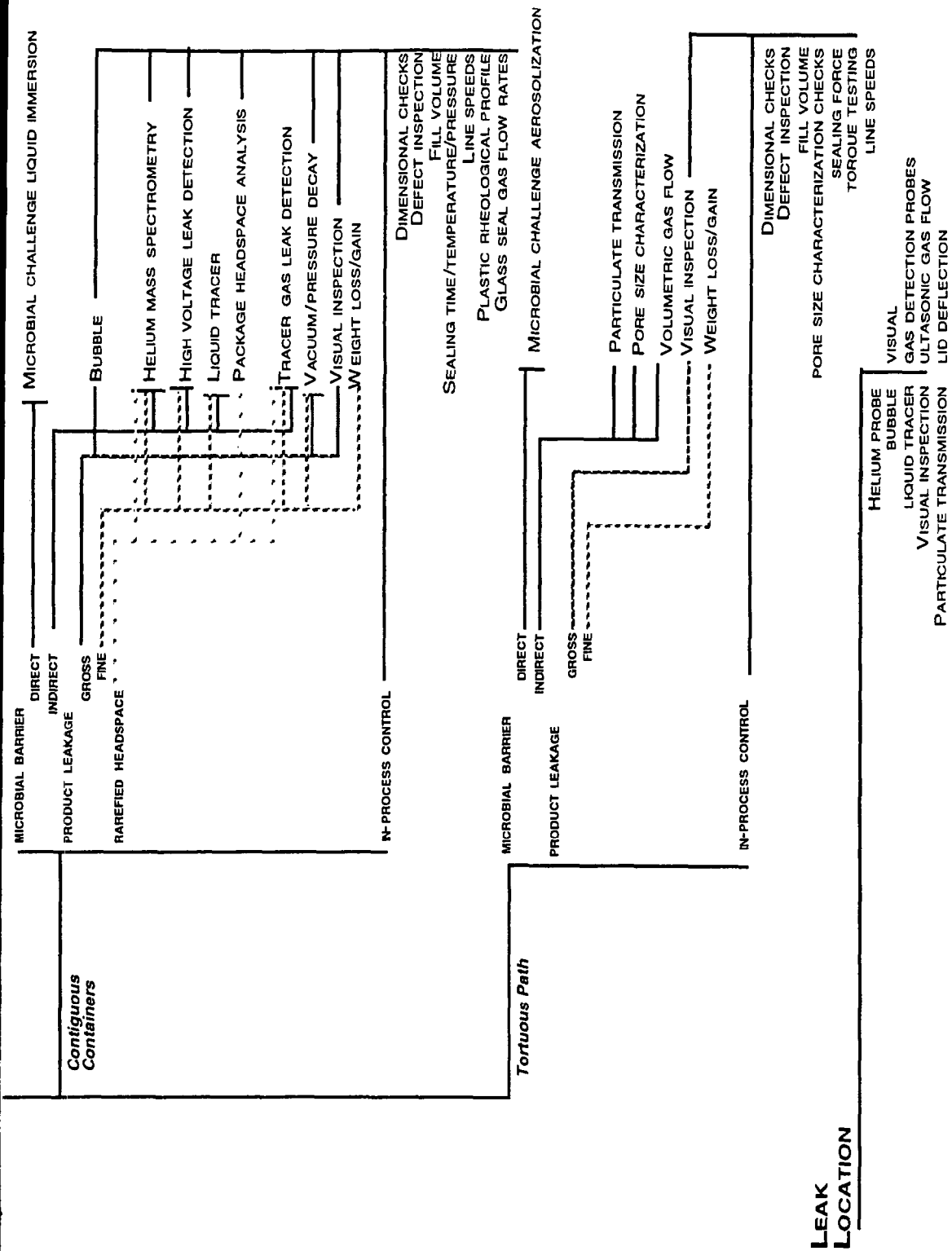
Below each performance criteria in the decision tree, a listing of potentially useful leak test methods is presented. Leak test methods are categorized according to their most common application. Microbial barrier tests are either described as direct (microbial challenge methods) or indirect (physical tests). Product leakage tests are subdivided into either gross methods (for larger leaks) or fine (for relatively smaller leaks). Rarefied gas headspace retention tests and vacuum retention tests are listed. Leak tests useful as in-process control checks are also provided. All leak tests included in the decision tree are described more fully in Section 8, Leak Test Methods. Certainly this decision tree is merely intended as a guide, and is not meant to be restrictive when choosing among these or other viable leak test methods or process controls.

Decision Tree Table of Terms

Leakage characterization	Total leakage and permeation	This refers to the need to detect, qualitatively or quantitatively, the total leakage and permeation
	Leak location	In this case, the source of leak in the container system is sought.
Package Seal Type	Compression or friction fit closure	A closure mechanism effected by the physical contact of two materials forced against each other, such as a rubber stopper on a glass vial.
	Thermal/adhesive bond	A bond formed by thermal fusion of two materials (e.g., a plastic laminate heat seal) or by the use of an intermediate adhesive.
	Contiguous container	A package formed from a single material (e.g., a glass or plastic ampoule)
	Tortuous path dependent closures	A barrier to airborne microorganisms accomplished primarily via a convoluted pathway to the product (e.g., a crimp-sealed aluminum tube, or a screw-thread closure)
Performance criteria	Microbial barrier	The selected method is used to demonstrate that the package system is an effective barrier against microbial ingress. In the case of physical detection methods, correlation between microbial ingress and the physical method is needed.
	Product leakage	The method is used to demonstrate that product is contained by the package.
	Rarefied gas atmosphere	The method is used to demonstrate that a controlled environment is maintained in the package headspace.
	Vacuum retention	The method demonstrates the maintenance of a vacuum in the package headspace.
	In-process control	The selected leak test or process control method is used to demonstrate that a manufacturing unit operation or series of operations is "in control." A correlation must exist between process control and product quality.
In-process parameter controls		These are manufacturing parameters which can be controlled to increase the probability of satisfactory package integrity

Decision Tree for the Selection of Potentially Useful Pharmaceutical Container/Closure Integrity Methods




Contiguous Containers
MICROBIAL BARRIER
MICROBIAL CHALLENGE LIQUID IMMERSION
PRODUCT LEAKAGE
DIRECT
INDIRECT
RAREFIED HEADSPACE
GROSS
FINE
BUBBLE
HELIUM MASS SPECTROMETRY
HIGH VOLTAGE LEAK DETECTION
LIQUID TRACER
PACKAGE HEADSPACE ANALYSIS
TRACER GAS LEAK DETECTION
VACUUM/PRESSURE DECAY
VISUAL INSPECTION
WEIGHT LOSS/GAIN
IN-PROCESS CONTROL
DIMENSIONAL CHECKS
DEFECT INSPECTION
FILL VOLUME
SEALING TIME/TEMPERATURE/PRESSURE
LINE SPEEDS
PLASTIC RHEOLOGICAL PROFILE
GLASS SEAL GAS FLOW RATES
Tortuous Path
MICROBIAL BARRIER
MICROBIAL CHALLENGE AEROSOLIZATION
PRODUCT LEAKAGE
DIRECT
INDIRECT
GROSS
FINE
PARTICULATE TRANSMISSION
PORE SIZE CHARACTERIZATION
VOLUMETRIC GAS FLOW
VISUAL INSPECTION
WEIGHT LOSS/GAIN
IN-PROCESS CONTROL
DIMENSIONAL CHECKS
DEFECT INSPECTION
FILL VOLUME
PORE SIZE CHARACTERIZATION CHECKS
SEALING FORCE
TORQUE TESTING
LINE SPEEDS
LEAK LOCATION
HELIUM PROBE
BUBBLE
LIQUID TRACER
VISUAL INSPECTION
PARTICULATE TRANSMISSION
VISUAL
GAS DETECTION PROBES
ULTASONIC GAS FLOW
LID DEFLECTION

8. Leak Test Methods

The final section is a series of summaries that describe several commonly used microbial and physical integrity test methods. These methods are listed alphabetically. In each summary, information is included on the validatability of the method, the limitations that influence the method's sensitivity, and whether it is useful for quantitative leakage measurement, for qualitative leakage detection, or for leak location. Critical test parameters are listed followed by the time required to complete the leak test, and special equipment required. The package stress created by the leak test is stated, along with specific product/package types that are most commonly tested by the given method. Finally, pertinent literature references from the document's general reference list are given that relate to the method being summarized.

8.1 Bubble Test

Description: The package is submerged in water or other suitable clear, colorless solvent. A vacuum is exerted on the test system, and the package is examined visually for evidence of gaseous leakage. Alternatively, surfactant may be added to the seal in question, a differential pressure is applied across the seal, and evidence of gaseous leakage is detected by visible inspection.

Validation	This method determines the presence and location of leaks that allow sufficient gaseous flow to permit the formation of visually detectable bubbles in the submersion fluid or in the applied surfactant. This method can be validated for the detection or location of larger leaks or package imperfections. Validation will depend on the test method parameters and the reliability and sensitivity of visual detection.
Sensitivity	The sensitivity of the method is based on the ability of gas to move out of a leak to form visible bubbles in the submersion fluid or applied surfactant, and the ability to identify positive leak controls
Quantitative/Qualitative	Qualitative
Critical test parameters	Differential pressure exposure Differential pressure rate of exposure Test time Submersion fluid or surfactant surface tension Cleanliness of package Visible inspection parameters include: • Light intensity, • Magnification, • Background, • Time for inspection • Visual inspector training and experience
Time to complete	Up to several minutes
Equipment required	Vacuum or pressure source Submersion fluid or surfactant Submersion vacuum vessel for submersion testing Pressure controls and monitors Visual inspection requires appropriate lighting, light meters, magnification, and color background.

Stress on package	Pressure differential Wetting of seal or entire package	
Product/Package Types	Applicable to all pharmaceutical packages that do not rely on a tortuous path or barrier seal. And that can tolerate wetting and pressure differentials	
Literature References	ASNT, 1982 ASTM D 3078 ASTM D 4991 ASTM E 515	Elder, 1995 Guazzo, 1996 Morton, 1987 Morton et al, May-June 1989

8.2 Helium Mass Spectrometry

Description: Helium is used as a tracer gas for detection and measurement of leakage across a package seal. Helium is detected by mass spectrometry. There are four ways of testing for helium leakage rate by mass spectrometry: Tracer Probe, Hood, Detector Probe, and Inside Out.

- Tracer Probe: The package itself is placed in-line with the mass spectrometry unit and helium is sprayed around the outside of the package. The mass spectrometer vacuum pump applies vacuum to the package and helium leakage into the package is measured by mass spectrometry.

- Hood Method: The package is placed in-line with the mass spectrometer and is enclosed in a "hood" or chamber into which helium can be introduced. The mass spectrometer vacuum pump applies vacuum to the package and helium leakage into the package is measured by mass spectrometry.

- Detector Probe: The package is flooded with helium and a detector probe connected to the mass spectrometer is used to sample the atmosphere around the package perimeter. The package is not under vacuum in this case.

- Inside-Out: The package is flooded with helium and subsequently placed into a test chamber directly connected to the mass spectrometer. Helium can be introduced into the package headspace by penetrating the package and flooding the headspace, by flooding the headspace prior to sealing the package, or by bombing, i.e. exposing the sealed package to an environment of pressurized helium. During the Inside-Out method the package is subjected to vacuum generated by the mass spectrometer.

Validation	Helium mass spectrometry methods can be validated for the detection, location, and measurement of gaseous leaks and helium permeation. Instrument sensitivity can be demonstrated using calibrated helium leaks. Positive control package leaks can be used in some cases. Validation studies should examine the ability of the leak test method to accurately measure gas leakage in the package system.
Sensitivity	Helium leak detection methods are potentially the most sensitive indicators of package leakage and permeation. The sensitivity of helium mass spectrometry methods is based on the ability to accurately detect and measure helium leakage/permeation rate in the package system. The test method and the parameters selected for testing the package are critical for accurate leakage/permeation measurement. General mass spectrometry system parameters that can impact sensitivity include: • Volume of vacuum system • Pressure in the vacuum system • Pressure in the sensing element of the mass spectrometer • Response time of the mass spectrometer • Effective pump speed of vacuum pump system • Ratio of the total gas flow of the vacuum pump system to the total gas flow of the mass spectrophotometer leak detector.
Quantitative/Qualitative	Quantitative: Hood method and Inside-Out method Qualitative or semi-Quantitative: Tracer Probe method and the Detector Probe method
Critical test parameters (all methods)	Package testing mode Test time duration Temperature of package Percentage of tracer gas surrounding the test boundary Location, length and size of connection between tester and vacuum system Helium background within the vacuum system Cleanliness, surface area, surface finish and material composition of test boundary exposed to the vacuum system Stability of electrical supply to the mass spectrometer

Critical test parameters (Hood and Inside-Out methods)	Concentration of the tracer gas in the package or hood Bombing time Bombing pressure Degassing of the package surface (post bombing dwell time or air wash)
Critical test parameters (Tracer and Detector Probe methods)	Specialized Operator training, experience, technique Surface scanning speed Differential pressure across the package Helium gas concentration within the test boundary The distance the probe or sniffer is held from the surface The length and diameter of sniffer probe lines Pressure in the sensing element of the mass spectrometer Build-up of helium in the background atmosphere
Time to complete	Method dependent. Several hours may be required for bombing. Once helium challenge conditions are established, the actual test time is a few seconds.
Equipment required	Helium mass spectrometer with vacuum system Package test fixture or probe Helium source
Stress on package	Vacuum (Tracer Probe, Hood and Inside-Out methods)
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier material and that are required to maintain a very low leak rate specification. Generally, the Inside-Out method is used for testing hermetically sealed containers having a defined internal volume that can withstand a vacuum applied to the package exterior. The Tracer Probe method is used for items that can withstand a vacuum of 0.1 Pascal (10^{-3} torr). The Detector Probe method is used for items that can tolerate a pressure differential and which the effluent side of the leak is accessible with the mass spectrometer probe. The Hood method can be used with packages that can be evacuated.

Literature References	ASNT 1982	Gilchrist, et al, 1985
	ASTM E 493	Gilchrist et al, 1989
	ASTM E 499	Guazzo, 1994
	ASTM E 908	Kirsch et al, 1997
	ASTM E 948	Hoffmann, 1996
	ASTM E 1603	Rhea, et al, 1984
	Elder, 1995	Varian Vacuum Products, 1995

8.3 High Voltage Leak Detection (HVLD)

Description: Containers are exposed to a high frequency, high-voltage current near the area of the package to be detected for leaks. The presence of a leak will result in a flow of current above base noise detected by the HVLD system as a reject. This test method is only feasible for containers constructed of relatively nonconductive materials and for product that is significantly more conductive. The product must be at or very near any potential leaks to allow conductance to occur.

Validation	HVLD methods can be validated for the detection of leaks in product/package systems meeting the conductivity requirements stated above. Validation studies should examine the ability of HVLD to detect control leaks as a function of voltage level and detection sensitivity settings, HVLD probe positioning, package wall thickness and geometry, and product formulation variations.	
Sensitivity	HVLD can detect leaks large enough to permit liquid penetration or wetting. HVLD sensitivity must be demonstrated during validation studies for each product/package system to ensure the detection of all control leaks with a minimum number of false positives.	
Quantitative/Qualitative	Qualitative	
Critical test parameters	High voltage setting Sensitivity setting (Gain) HVLD probe positioning Package geometry Package conductivity	Product conductivity Test area cleanliness Cleanliness of the package Access of product to the leak
Time to complete	Test time is almost instantaneous.	

Equipment required	High voltage leak detection instrumentation Package handling system to ensure package is positioned to wet the seal area and for feeding product into the HVLD equipment.
Stress on package	Rapid, short exposure to a voltage.
Other	It is recommended that the impact of voltage on product stability be determined.
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal, and that meet the conductivity requirements described above.
Literature References	Anonymous, 1996 Arndt, et al, 1996 Axelson, et al, 1990 Ball, 1997 The Parenteral Society, 1992 Villiger 1987

8.4 Lid Deflection

Description: To perform lid deflection tests a pressure differential is applied to a package and the resultant movement of the package's flexible lidding material is monitored and used as an indication of the absence or presence of unacceptable levels of leakage. Lid deflection sensors can work by any of a variety of methods, including strain gauges for monitoring lidding movement, and sensors that work by contact, capacitance, inductance, electromagnetic wave motion, photoelectricity, and ultrasonic pulse echo.

Validation	Lid deflection methods can be validated for the detection of the absence of lidding material or for the detection of gaseous leaks large enough to result in minimal or no flexible lidding material displacement when a package is subjected to a reference differential pressure. Validation studies should examine the ability of the lid deflection instrument to detect control leaks as a function of instrument sensor capabilities, differential pressure testing conditions, package headspace, and package component geometry.

Sensitivity	The sensitivity of lid deflection methods is based on the ability to accurately detect flexible lidding material movement when the package is subjected to a standard differential pressure. Sensitivity will largely depend on the method for monitoring lidding movement and the capabilities of the particular sensor device selected. Instrument sensor capabilities must be verified using appropriate techniques specific for the sensor selected. Generally, lid deflection tests are used as an indicator of larger leaks. This method is sometimes coupled with pressure/vacuum decay leak tests in order to provide both a "gross" and a "fine" leak test detection range.
Quantitative/Qualitative	Qualitative
Critical test parameters	Lidding movement sensor accuracy and sensitivity Differential pressure applied to package Package headspace Package component geometry
Time to complete	A few seconds
Equipment required	Lidding deflection instrumentation Differential pressure source, monitors and controls
Stress on package	Differential pressure
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal and that can tolerate a differential pressure applied to the seal area. This test is particularly useful for checking critical sizes of specialty packages or devices. The test can often be used instead of vacuum/pressure decay for packages that tend to deform under pressure.
Literature References	Gnanasekharan, et al, 1992 Gnanasekharan, et al, 1994 Guazzo, 1996 Hoffmann, 1996 Rice, 1986 Yam, 1995

8.5 Liquid Tracer

Description: A liquid tracer solution, such as dye, radionuclides, or metal ions, is exposed to the package seal to be tested. This may require submersion of the package in the liquid tracer, application of a small amount of liquid tracer to the seal, or filling the package itself with liquid tracer followed by submersion of the package in water or other suitable solvent. Migration of the tracer across the seal is determined either visually, spectrophotometrically, or by other analytical

means appropriate to the liquid tracer. Often a pressure differential is applied to the package to accelerate the migration of the tracer across the seal and to improve test method sensitivity.

Validation	Liquid tracer leak test methods can be validated for the detection or location of leaks large enough to permit liquid penetration and detection. For some methods, the extent of liquid leakage can be quantitatively measured. Validation will depend on the test method parameters and the reliability and sensitivity of the liquid tracer detection method.
Sensitivity	The sensitivity of the method is based on the extent of liquid tracer penetration and the ability to detect or measure such leakage.
Quantitative/Qualitative	Generally qualitative, although it may be quantitative if liquid tracer penetration can be quantitatively measured.
Detection limit	There must be sufficient liquid tracer leakage to permit detection by the selected detection method.
Critical test parameters	General test parameters include: • Differential pressure exposure • Time of liquid tracer exposure • Tracer liquid surface tension, viscosity, hydrophilicity/lypophilicity, and temperature • Compatibility and stability of the liquid tracer with the product • Light stability of the tracer with product Visual inspection parameters include: • Type of lighting and light intensity • Magnification • Background • Angle of inspection • Time for inspection • Training and experience of inspectors Analytical inspection parameters will depend on the detection method of choice.
Time to complete	Several minutes to as long as a few hours

Equipment required	The equipment required varies with the liquid tracer and the detection method selected. Some equipment that may be required includes: Liquid tracer solution Vacuum or pressure source Submersion fluid Vacuum vessel for submersion testing Pressure controls and monitors Visual inspection requires appropriate lighting, light meters, magnification, and color background. Fluorescent dye inspection requires black light of appropriate wavelength and intensity Analytical detection method may require UV/Vis spectrophotometry, atomic absorption, HPLC, or other appropriate instrumentation	
Stress on package	Pressure differential Wetting of seal or entire package	
Product/Package Types	Applicable to all pharmaceutical packages that do not rely on a tortuous path or barrier seal and that can tolerate wetting or submersion	
Literature References (general)	ASNT Handbook, 1982 ASTM E 165 Elder, 1995 Guazzo, 1994 Guazzo, 1996 Morton, 1987	McVean et al, 1972 PDA Techn. Inform. Bull. 4, 1983 Korczynski 1987
Literature References (dye tests)	Akers, 1985 Artz et al, 1961 ASTM E 165 British Standards Institute, B.S. 3263, current edition Chrai et al, Sep 1994 Gilchrist et al, 1989 EP VI	Hansen et al, 1995 ISO 8362-2 Jones et al, 1995 Lin, et al, 1982 Marcy, 1995 McVean et al, 1972 Spitzley, 1993 The Parenteral Society, 1992
Literature References (other)	Butler et al, 1978 [radionuclides]	Morton et al, May-June 1989 [copper ion]

8.6 Microbial Challenge, Aerosolization

Description: Microbial challenge testing by aerosolized microorganisms involves placing product-filled or media-filled package units into a specially designed aerosolization chamber and charging the atmosphere within the chamber with a nebulized cloud or aerosol of microorganisms for a specified period of time. During the challenge test, the packages may be subjected to differential pressure challenges. The atmosphere within the chamber must be kept within defined humidity and temperature ranges and well circulated. After the challenge, the units are removed from the chamber. If media-filled product is used (or product which supports microbial growth), the units are then incubated and examined for microbial growth. Otherwise, the package contents are removed and exposed to media which is subsequently incubated and examined for growth.

Validation	Microbial aerosolization challenge methods determine the presence of microorganisms that have penetrated a package exposed to aerosolized microorganisms. Critical validation parameters include the concentration and uniformity of aerosolized microorganisms, chamber temperature, pressure, and humidity settings and controls, and the ability of the media or product to support microbial growth. Positive controls should demonstrate growth.
Sensitivity	The sensitivity of the method is based on the incidence of ingress of aerosolized microorganisms into positive control package defects and the ability to detect such ingress.
Quantitative/Qualitative	Qualitative
Detection limit	There must be sufficient bacterial growth for media to visibly or analytically demonstrate growth.
Critical test parameters	Media or product microbial growth promotion Stable microbial concentration in the aerosol (level and uniformity) Size of challenge organism Electrical charge on aerosolized particle Extent of microbial drying during test Time of exposure to the aerosol Chamber air flow patterns and speed Chamber temperature and humidity conditions Differential pressure exposure (optional)
Time to complete	Exposure time is typically a few hours. Incubation time is several days, depending on the microbial challenge and incubation conditions

Equipment required	Standard microbiological laboratory equipment Microbial growth support media or product Airtight test chamber Microbial spore nebulization unit with impinger, motor and fan Temperature, pressure, and humidity controls and monitors Chamber sterilization equipment Air samplers Standard microbiological laboratory equipment
Stress on package	Pressure differential (optional)
Challenge Organism	Typically, *B. subtilis* spores
Product/Package Types	Applicable to sterile pharmaceutical packages that cannot tolerate liquid immersion microbial challenge tests and that rely on tortuous path or barrier seals to ensure package integrity
Literature References	Akers, 1985 ASTM F1585-95 ASTM F1608-95 Blakistone, 1995 Chen, et al, 1991 Chrai, et al Sept. 1994 Gnasasekharan, et al, 1995 Hansen, et al, 1995 Henke, et al, 1992 Jones, et al, 1995 Korczynski, 1987 Morton, 1987 Placencia, et al, 1988, 1989 Reich, 1985 Reich, 1986 Spitz, 1994 Spitzley, 1993 Tallentire, et al, 1985

8.7 Microbial Challenge, Liquid Immersion

Description: Microbial challenge testing by liquid immersion involves immersing filled package units into a liquid concentration of microorganisms for a specified period of time, removing the units, rinsing them, incubating the units and examining them for microbial growth. These units typically contain nutritive growth media, but may alternatively contain product, if not inhibitory to the growth of the challenge organism. During immersion, the packages may be exposed to extremes of pressure to simulate anticipated product processing or distribution conditions.

Validation	Microbial challenge by liquid immersion tests determine the ingress of microorganisms into a package that has been challenged with a liquid microbial challenge suspension. A method can be validated, provided the concentration of microorganisms in the challenge suspension is sufficient, and the media or product within the container adequately supports microbial growth. Positive controls, if appropriate, should demonstrate growth.
Sensitivity	The sensitivity of the method is based on the incidence of ingress of liquid-borne microorganisms through positive control package defects and the ability to detect such ingress. Packaged media or product must support microbial growth.
Quantitative/Qualitative	Qualitative
Detection limit	There must be sufficient bacterial growth for media to be visibly turbid or analytically detectable
Critical test parameters	Challenge media formulation and surface tension Size of challenge organism High viable count in challenge media at end of test ($\geq 10^5$/mL) Time of exposure to microorganism Differential pressure exposure Ability of sterile packaged media to support growth of challenge organism. Sensitivity of analytical method if used to detect growth
Time to complete	Exposure time can vary from a few minutes to several hours. Usual incubation time is 7-14 days
Equipment required	Standard microbiological laboratory equipment Microbial growth support media or product Challenge microorganism Immersion chamber capable of vacuum/pressure cycles and temperature control Incubation chamber For visual inspection, an inspection area with appropriate lighting and color background For analytical microbial growth determination, appropriate equipment and instrumentation
Stress on package	Pressure differential Wetting of seals from liquid media

Challenge Organism	Organisms reportedly used include: *Escherichia coli, Serratia marcesans, Clostridium sporogenes, Pseudomonas aeruginosa, Staphylococcus epidermidis, Brevundimonas diminuta*
Product/Package Types	Applicable to sterile pharmaceutical packages that can tolerate liquid immersion
Literature References	Akers, 1985 Blakistone, 1995 Chrai, et al, Sept. 1994 Frieben, et al, 1982 Harper, 1995 Kirsch, et al, 1997 Korczynski, 1987 Gnanasekharan, et al, 1995 Lin, et al, 1992 Marcy, 1995 McEldowney, et al, 1990 Morton, 1987 Morton, May/June 1989 PDA, 1983 Rose, 1995 The Parenteral Society, 1992

8.8 Package Headspace Analysis

Description: The headspace content of a package is analyzed and monitored for change over time. Package headspace analyses are useful for the measurement of leakage and permeation from packages that require a rarefied gas headspace to ensure long term product stability. The packages may be stored in a challenge environment of vacuum or concentrated rarefied gas to accelerate leakage/permeation.

Validation	This method can be validated for the measurement of gaseous leaks and permeation large enough to result in an unacceptable shift in headspace content over time. Validation studies should examine the ability of the analytical method to accurately sample and measure rarefied gas content in the package system.
Sensitivity	The sensitivity of headspace analyses is based on the ability to accurately sample and measure gas headspace content in the package system. Sensitivity can be verified using reference gas mixtures of known concentration. Gas headspace content tests can be very sensitive indicators of package leakage and permeation, especially after exposure of product to longer times or to a challenge environment designed to accelerate leakage and permeation.
Quantitative/Qualitative	Quantitative

Critical test parameters	Rarefied gas analytical instrument sensitivity and accuracy Headspace sampling technique Package headspace volume Efficiency of the establishment of initial gas headspace content
Time to complete	Test times can be a few seconds to a few minutes
Equipment required	Analytical gas content measurement instrumentation Headspace sampling device
Stress on package	Puncture during headspace sampling
Product/Package Types	Applicable to pharmaceutical packages that require a rarefied gas headspace to ensure product stability.
Literature References	ASNT, 1982 Wang, et al, 1997

8.9 Particulate Transmission

Description: Package challenge testing by aerosolized dust involves placing the package units into a specially designed aerosolization chamber and charging the atmosphere within the chamber with aerosolized dust for a specified period of time. During the challenge test, the packages may be subjected to differential pressure challenges. The atmosphere within the chamber must be kept well circulated. After the challenge, the units are removed from the chamber and examined for evidence of dust ingress.

Validation	Particulate transmission tests determine the presence and location of leaks that can be penetrated by particulates. This method can be validated provided the particulate size is uniformly small, the dust concentration is sufficient and uniformly dispersed within the chamber, and the chamber circulation system settings and controls are adequate. Positive controls, where appropriate, should demonstrate dust ingress.
Sensitivity	The sensitivity of the method is based on the incidence of penetration of aerosolized dust into positive control package defects and the ability to detect such penetration.
Quantitative/Qualitative	Qualitative
Detection limit	There must be sufficient dust for visible detection.

Critical test parameters	Size and uniformity of challenge dust particle Concentration of dust in aerosol Electrical charge on dust Time of exposure to dust Chamber air flow patterns and speed Chamber temperature and humidity conditions Differential pressure exposure (optional)
Time to complete	Exposure time is typically a few hours.
Equipment required	Particulate source Airtight test chamber Dust aerosolization unit with motor and fan Pressure controls and monitors
Stress on package	Pressure differential (optional)
Challenge Tracer Dust	Carbon black, talcum powder, powdered chalk, smoke
Product/Package Types	Applicable to sterile pharmaceutical packages that rely on a tortuous path or barrier system.
Literature References	Jones, et al, 1995

8.10 Residual Gas Ionization

Description: A high voltage, high frequency field is applied to packages sealed under vacuum. The field causes residual headspace gas to ionize. A conducting plate located near the package detects the level of ionization current. Ionization above an established limit triggers rejection of the package.

Validation	The residual gas ionization method can be validated for the detection of insufficient vacuum in packages designed to maintain a minimal vacuum level. Validation studies should examine the ability of the ionization sensors to differentiate between packages containing various known levels of vacuum as a function of voltage, frequency, or package headspace.

Sensitivity	The sensitivity of a residual gas ionization method is based on the ability to accurately detect pressure exceeding a desired acceptance limit in evacuated packages. Instrument sensitivity can be determined using control packages containing known pressure levels. It is possible to detect smaller leaks and permeation rates with this method as the time interval from when the package was originally evacuated to the test time is lengthened.
Quantitative/Qualitative	Qualitative
Critical test parameters	Voltage Frequency Conducting plate and voltage field positioning Package component ionization potential Package headspace volume Gaseous content of the residual headspace Package component geometry
Time to complete	Less than a second.
Equipment required	Gas ionization instruments Package handling systems
Stress on package	High frequency high voltage spark
Product/Package Types	Applicable to pharmaceutical packages designed to maintain a partial vacuum in the headspace. The package tested must be relatively non-ionizing. Impact of the voltage spark on product stability should be considered.
Literature References	Wu, 1989 Crist, 1994

8.11 Tracer Gas Leak Detection

Description: A tracer gas is placed on one side of the package seal. An inert carrier gas is passed across the opposite seal side. Tracer gas is detected by appropriate analytical methods. Examples of analytical methods include coulometric detectors for oxygen detection and photoelectric sensors for carbon dioxide and water vapor detection. This test is useful for the measurement of leakage and permeation from packages that require a specific gas headspace to ensure long term product stability, or for products that are susceptible to degradation upon exposure to environmental gases or vapors. While oxygen, carbon dioxide, and water vapor are

typically selected for testing pharmaceutical packages, a wide variety of other tracer gases can be used as well (see Ref. Elder, 1995; ASNT, 1982).

Validation	This method can be validated for the detection, location, and measurement of gaseous leaks and permeation large enough to result in an unacceptable shift in headspace content over time or in product degradation. Validation studies should examine the ability of the analytical method to accurately measure gas leakage/permeation rate in the package system.
Sensitivity	The sensitivity of tracer gas detection methods is based on the ability to accurately measure tracer gas leakage/permeation rate in the package system. Instrument sensitivity can be established using reference gas mixtures of known tracer concentration. Test method sensitivity is largely dependent on the design of the test fixture.
Quantitative/Qualitative	Quantitative
Critical test parameters	Gas analytical instrument sensitivity and accuracy Package test fixture design and dimensions Temperature and humidity of the testing environment Inert carrier gas temperature and humidity Concentration of the tracer gas in the package Concentration of tracer gas in the effluent atmosphere
Time to complete	Several hours to a few days
Equipment required	Analytical gas measurement instrumentation with package test fixture. Carrier gas source Tracer gas
Stress on package	None
Product/Package Types	Applicable to pharmaceutical packages that are required to maintain a specified leakage/permeation rate of gas through the package to ensure product stability.
Literature References	Amini, 1981 ASNT, 1982 ASTM E 427 ASTM E 498 ASTM E 1603 Demorest, 1988 Demorest, 1992 Demorest, et al, 1996 Elder, 1995 Gnasasekharan, 1994 Guazzo, 1996

8.12 Ultrasonic Detection of Gas Flow

Description: Containers under pressure are placed near an ultrasonic sensor capable of detecting acoustic energy in the ultrasonic range of 20-100 kHz. Ultrasonic sensors will detect movement of air or gasses out of package leaks large enough to create turbulent gas flow. Smaller leaks may be detected by the application of a special surfactant to the seal. Bursting of surfactant bubbles is detected ultrasonically. There must be enough package headspace to permit detectable gas flow out of the package.

Validation	Ultrasonic gas flow detection methods can be validated for the detection and location of leaks in package systems sealed under pressure or in packages which can be "squeezed" or otherwise manipulated to create pressure inside the package. Validation studies should examine the ability of the ultrasonic sensor to detect control leaks as a function of instrument sensitivity settings, internal package pressure, package-sensor positioning, background noise, and line speeds.	
Sensitivity	Sensitivity of the method is limited by the presence of turbulent flow gas leakage and the ability to detect such flow using positive control package leaks or standard gaseous reference leaks.	
Quantitative/Qualitative	Qualitative	
Critical test parameters	Sensitivity settings Internal package pressure Package-sensor positioning	Background noise Line speeds
Time to complete	Test time is almost instantaneous.	
Equipment required	Ultrasonic detection instrumentation Package handling system for squeezing package and/or positioning of the package near the sensor. Surfactant for ultrasonic detection of bubble breakage	
Stress on package	Possible squeezing of flexible packages Possible surfactant wetting of seal surface	
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal, that contain a headspace volume, that are sealed under pressure or that are flexible enough to permit pressurization by squeezing.	

Literature References	ASNT, 1982 ASTM E 1002 Bandes, 1988 Elder, 1995 Guazzo, 1997

8.13 Ultrasonic Imaging

Description: Very short ultrasound pulses are generated at frequencies up to several hundred MHz. These ultrasound frequencies are focused on the container seal by acoustic lenses. Once inside the package the ultrasonic pulses may be absorbed, scattered or reflected depending on the change in the material. The reflected ultrasound echoes are used to create an image that can be analyzed for the presence of improper sealing, delamination, gaps, or product-contaminated seals. Alternate ultrasonic inspection methods work by contact transmission of ultrasonic energy through the package seal using an array of transmitters and receivers in direct contact with the seal region.

Validation	Ultrasonic imaging methods can be validated for the detection and location of seal imperfections in flat thermal or adhesive bonded seals, for instance a pouch. Validation studies should examine the ability of the ultrasonic sensor to detect control leaks as a function of ultrasound pulse generation and transmission variables, instrument sensor sensitivity settings, transmission and detection sensor positioning, package materials of construction, and package geometry. Validation studies should reflect the range of acceptable package sealing parameters.
Sensitivity	Sensitivity of an ultrasonic imaging method is based on the ability to detect positive control package defects such as gaps, seal contaminants, package material defects, and other seal inconsistencies.
Quantitative/Qualitative	Qualitative
Critical test parameters	Sensor sensitivity settings Quantity of transmitters and sensors per seal surface area Package-sensor and transmission positionings Ultrasound frequency Ultrasound pulse amplitude Package materials of construction Package geometry Dry package testing vs. water immersion testing

Time to complete	Test time is less than one second
Equipment required	Ultrasonic detection and imaging instrumentation Package handling system for positioning packages in the sensor.
Stress on package	None
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal, that contain a headspace volume, that are sealed under pressure or that are flexible enough to permit pressurization by squeezing.
Literature References	Adams, 1995

8.14 Vacuum/Pressure Decay

Description: A pressure differential is applied to a package and the change in pressure or vacuum is measured either inside the package itself using a test fixture, or outside in a sealed package test chamber. The change in test chamber pressure or vacuum can be determined by absolute pressure measurement, or it can be determined using a differential pressure manometer in which the test chamber pressure change is compared to that of a control package.

Validation	Vacuum/pressure decay methods can be validated for the detection or measurement of gaseous leaks large enough to allow a measurable change in pressure in the package or the test chamber over the duration of the test. Validation studies should examine the ability of the vacuum/pressure decay instrument to detect control leaks as a function of test chamber geometry, instrument sensor capabilities, differential pressure testing conditions, test time, and environmental temperature conditions.
Sensitivity	Sensitivity of the vacuum/pressure decay method is based on the ability to detect and/or measure a change in pressure or vacuum resulting from gaseous leaks. Instrument sensitivity can be verified using positive control package defects or gaseous reference leaks. Test method sensitivity is greatly influenced by a number of factors. These include those mentioned under validation above as well as the critical parameters listed below. Generally, pressure decay is considered a more sensitive test that vacuum rise, and differential pressure is potentially more sensitive than absolute pressure sensors.

Quantitative/Qualitative	Qualitative or quantitative, depending on the instruments and test conditions used.
Critical test parameters	Pressure/vacuum sensor mechanism of operation and capabilities Volume of space being tested (test chamber or package headspace) Test chamber geometry Time allowed for measured pressure change to occur Stabilization time Differential pressure applied to package Water vapor pressure inside package Atmospheric pressure Temperature of package and test system Package creep effects when exposed to differential pressure
Time to complete	Test time may range from less than a second to a few minutes.
Equipment required	Leak test instrumentation including vacuum or pressure sensors in absolute or differential pressure sensing mode Differential pressure source, monitors and controls Temperature and humidity controls and monitors
Stress on package	Differential pressure
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal and that can tolerate a differential pressure applied to the seal area. When selecting the appropriate testing configuration and parameters, consider the impact of package headspace volume (vacuum testing requires a defined headspace), and the presence of product which may plug the leaks (vacuum testing may pull liquids, semi-solids, or even moisture in the package into the seal area thus plugging leaks), and the ability of the package to tolerate differential pressure (vacuum rise tests are limited to a differential pressure of 1 atm whereas pressure decay differential pressure may be much greater).

Literature References	ASNT, 1982 Blakistone, 1995 Chrai, et al, 1994 Elder, 1995 Gnanasekharan, et al, 1992 Gnanasekharan, et al, 1994 Guazzo, 1994 Guazzo, 1996	Hoffmann, 1996 Jones, et al, 1995 Morton, 1987 Morton, et al, Mar-Apr 1989 Morton, et al, May-Jun 1989 Snow, 1992 Stauffer, 1988 The Parenteral Society, 1985 Yam, 1995

8.15 Vacuum/Pressure Retention

Description: Vacuum/pressure retention tests consist of directly measuring the vacuum or pressure within a package headspace by typically using a package piercing device and an absolute pressure gauge. Vacuum/pressure retention tests are used for packages that are designed to maintain either a vacuum or positive pressure within the package as a function of product stability.

Validation	Vacuum/pressure retention methods can be validated for the detection or measurement of gaseous leaks large enough to allow a measurable change in pressure in the test chamber over the duration of the test. Validation studies should examine the ability to access the package headspace and accurately measure headspace pressure.
Sensitivity	Sensitivity of the vacuum/pressure retention method is based on the ability to detect and/or measure a change in pressure or vacuum resulting from gaseous leaks. Test method sensitivity is dependent on the ability to access the package headspace without creating a loss of pressure or vacuum.
Quantitative/Qualitative	Qualitative or quantitative
Critical test parameters	Pressure/vacuum sensor mechanism of operation and capabilities Volume of space being tested (test chamber or package headspace) Time allowed for measured pressure change to occur Vacuum or pressure initially introduced to the package Water or other vapor pressure inside package Temperature of package and test system Technique for piercing or accessing package headspace
Time to complete	Test time is a few seconds.

Equipment required	Absolute pressure gauge Temperature controls and monitors
Stress on package	Pressure or vacuum
Product/Package Types	Applicable to pharmaceutical packages that are designed to maintain a positive pressure or a vacuum in the headspace over product shelf-life.
Literature References	ASNT, 1982 Chrai, et al, 1994 Crist, 1994 Gnanasekharan, et al, 1992 Morton, 1987 PDA, 1981 Rhea, et al, 1984 Wang, et al, 1997

8.16 Visual Inspection

Description: The package is examined visually by human or by electronic inspection devices for evidence of product leakage. The test may be accompanied by additional package challenges of temperature or pressure extremes to increase the sensitivity of the test.

Validation	Visual inspection for leakage determines the presence or location of leaks penetrated by the contained product. This method can be validated for the detection of larger leaks or package imperfections. Validation will depend on the reliability and sensitivity of the mode of leakage detection.
Sensitivity	The sensitivity of the method is based on the incidence of product migration through positive control package defects and the ability to detect such migration.
Quantitative/Qualitative	Qualitative test
Detection limit	There must be sufficient product or contents leakage to permit visible or electronic detection.
Critical test parameters	Package cleanliness Differential pressure and temperature exposure (optional) Visible inspection parameters include light intensity, magnification, time, background, angle of inspection Visual inspector training and experience Electronic inspection parameters are dependent on the mode of inspection

Time to complete	From less than a second for electronic inspection to several minutes by visual inspection.
Equipment required	Temperature and/or pressure controls and monitors are required for optional package stress testing Visual inspection requires appropriate lighting, light meters, magnification, and color background. Electronic inspection requires appropriate product handling equipment and inspection instrumentation
Stress on package	Pressure or temperature differential (optional)
Product/Package Types	Applicable to all pharmaceutical packages and closure/seal types
Literature References	ASNT, 1982 ASTM E 165 Chrai, et al, 1994 Gnasasekharan, et al, 1992 Hansen, et al, 1995 Jones, et al, 1995 Lin, et al, 1982 Shaw, 1995

8.17 Volumetric Gas Flow

Description: A pressure differential is applied to a package and the outward flow rate of gas is measured.

Validation	Volumetric gas flow methods can be validated for the measurement of gaseous leaks large enough to allow a measurable flow of gas out of the package over the duration of the test. Validation studies should examine the ability of the volumetric flow rate instrument to measure control leaks as a function of instrument sensor capabilities, differential pressure testing conditions, and test time.
Sensitivity	The sensitivity of volumetric gas flow methods is based on the ability to accurately measure gaseous flow out of a package under positive pressure. Instrument sensitivity can be verified using gaseous reference leaks or positive control package defects. Test method sensitivity is influenced primarily by flow meter accuracy, test time, and the differential pressure applied to the package. Generally, volumetric gas flow is not as sensitive a test as vacuum / pressure decay methods.
Quantitative/Qualitative	Quantitative

Critical test parameters	Flow meter accuracy Time of test Differential pressure applied to package
Time to complete	A few seconds
Equipment required	Leak test instrumentation equipped with flow meters Differential pressure source, monitors and controls
Stress on package	Differential pressure
Product/Package Types	Applicable to pharmaceutical packages that do not rely on a tortuous path or barrier seal and that can tolerate a differential pressure applied to the seal area. This test is particularly useful for checking critical orifice sizes of specialty packages or devices. The test can often be used instead of vacuum/pressure decay for packages that tend to deform under pressure.
Literature References	ASNT, 1982 Gnanasekharan, et al, 1992 Gnanasekharan, et al, 1994 Guazzo, 1996 Hoffmann, 1996 Snow, 1992

8.18 Weight Loss/Gain

Description: The container is filled with either a volatile liquid or solid, or a hygroscopic material such as a desiccant. The container is stored in an environment intended to promote migration of the substances out of the package containing volatile materials, or to promote migration of water vapor into the package containing hygroscopic materials. Leakage plus permeation are measured in units of package weight loss or weight gain. For greater accuracy, it is recommended, although not required, that control packages filled with inert materials be tested at the same time as the test units. Each time the test units are weighed, the control packages' weight change should be subtracted from the change in test unit weights.

Validation	Weight loss or weight gain methods can be validated for the measurement of leaks of a wide variety of sizes, as well as for permeation measurement. Validation will depend on the test method parameters and the sensitivity of the balance used to monitor weight change. Acceptance limits for weight loss or gain are generally established as a function of the product-package system.

Sensitivity	The sensitivity of a weight loss/gain method is dependent on the extent of weight loss or gain. The weight change of the test units must vary significantly from the control units, and should be above the limit of balance sensitivity.
Quantitative/Qualitative	Quantitative
Critical test parameters	The volatility (weight loss) or hygroscopicity (weight gain) of the packaged material Time allowed for weight change Challenge environment conditions of temperature, and either pressure (weight loss) or relative humidity (weight loss or gain) Handling of samples and weighing techniques
Time to complete	A few minutes may be required for highly volatile materials (e.g., dry ice) Otherwise, several days or weeks may be required.
Equipment required	Balance For weight gain: • Relative humidity storage chamber • Storage chamber temperature and humidity monitors For weight loss: • Appropriate package storage environment with monitors • Optional: elevated temperature, differential pressure with appropriate monitors and controls
Stress on package	Environmental extremes of temperature, pressure, humidity.
Product/Package Types	Applicable to all pharmaceutical packages and seal types intended to contain products susceptible to weight change over shelf-life.
Literature References	Amini, 1981 Cilento, 1977 Guazzo, 1996 Morton, 1987 USP <661> USP <671>

9. Bibliography

Adams, T, "Hidden seal defects are target of new acoustic imaging technology," Food & Drug Packaging, April 1995.

Akers, M.J., Chapter 4: "Package Integrity Testing," Parenteral Quality Control: Sterility, Pyrogen, Particulate & Package Integrity Testing, 1st edition, M.J. Akers, ed., Marcel Dekker, NY, copy 1985.

Amini, M.A., "Testing permeation and leakage rates of pharmaceutical containers," Pharm. Technol., p. 39-43, Dec. 1981.

Amini, M.A., Morrow, D.R., "Leakage and permeation: Theory and practical applications," Package Development & Systems, p. 37-44, May/Jun 1979.

Am. Soc. For Nondestructive Testing (ASNT), Nondestructive Testing Handbook, Vol. 1., Leak Testing, 2nd ed., ASNT pub., Columbus OH, copy 1982.

Am. Soc. For Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103:

Testing Methods for:

D 3078	Leaks in heat sealed flexible pouches
D 4991-94	Leakage testing of empty containers by vacuum method
E 165-94	Liquid penetrant examination
E 493-94	Leaks using the mass spectrometer leak detector in the inside-out testing mode
E 498-94	Leaks using the mass spectrometer leak detector or the residual gas analyzer in the tracer probe mode
E 499-94	Leaks using the mass spectrometer leak detector in the detector probe mode
E 515-94	Leaks using bubble emission techniques
E 1002-94	Leaks using ultrasonics
E 1603-94	Leakage measurement using the mass spectrometer leak detector or residual gas analyzer in the hood mode
F1585-95	Integrity testing for porous barrier medical devices
F 1608-95	Microbial ranking of porous packaging materials (exposure chamber method)

Practices for:

E 427-94	Testing for leaks using the halogen leak detector (alkali-ion diode)
E 908-91	Calibrating gaseous reference leaks

Guides for:

E 432-91	Guide for the selection of a leak testing method
E 479-91	Guide for preparation of a leak testing specification
E 1916:	Terminology for nondestructive examination

Anonymous, "Ciba-Geigy evaluates faster ample test that may spot flaws better," Food & Drug Packaging, p. D-3, Jan 1996.

Arndt, Jr., G.W., Charbonneau, J., "Nondestructive electroconductivity testing of packages for hermetic integrity," Chapter 16, Plastic Package Integrity Testing. Assuring Seal Quality. B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

Artz, W.J., Gloor, Jr., W.T., Reese, D.R., "Study of various methods of detecting leaks in hermetically sealed ampules," J. Pharm. Sci., Vol. 50, p. 258-262, 1961.

Axelson, L., Cavlin, S., Nordstrom, J., "Aseptic integrity and microhole determination of packages by electrolytic conductance measurements," Pack. Technol. Sci., Vol 3: 141-162, 1990.

Ball, B., Yamanaka, A., "Validation of Nikka Densok Pinhole Inspector for BFS Applications," Blow/Fill/Seal News, Jan. 1997.

Bandes, A.S., "Leak detection methods through airborne ultrasound," J. Packaging Technol., Vol. 2, No. 3, p. 110-111, Jun 1988.

Blakistone, B., Harper, C., "New developments in seal integrity testing," Chapter 1, Plastic Package Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

British Standards Institute, B.S. 3263:1960, "Specification for rubber closures for injectable products," British Standards House, 2 Park St., London, UK, 1962.

Butler, L.D., Coupal, J.J., DeLuca, P.P., "The detection of ampul leakers using short-lived radionuclides," J. Parenter. Drug Assoc., Vol. 32, No. 1, p. 2-8, Jan/Feb 1978.

Chen, C., Harte, B., Lai, C., Pestka, J., Henyon, D., "Assessment of package integrity using a spray cabinet technique," J. Food Protection, Vol. 54, No. 8, p. 643-647, 1991.

Chrai, S., Heffernan, G., Myers, T., "Glass vial container-closure integrity testing—an overview," Pharm. Technol., p. 162-173, Sep 1994.

Cilento., R.D., "Methods for evaluating hermetic closure for screw-capped bottles," J. Pharm. Sci., Vol. 66, No., 3, p. 333-336, Mar 1977.

Connor, J.T., "In process verification of closure seal integrity," J. Parenter. Sci. Technol., Vol. 37, p. 14-19, 1983.

Crist, B., "Time-dependence of pressure in lyophilization vials," PDA J. Pharm. Sci. & Technol., Vol. 48, No. 4, p. 189-196, Jul/Aug 1994.

Demorest, R., "Nondestructive leak detection of blister packs and other sterile medical packages," J. Packaging Technol., Vol. 2, No. 5, Oct. 1988.

Demorest, R., "Recent developments in testing the permeability of good barriers," J. of Plastic Film & Sheeting, Vol. 8, Apr 1992.

Demorest, R., "Measurement of water vapor permeability: Recent developments and new standards," presented at Packforsk Symposium, Stockholm, Sweden, Mar 23, 1995 (available through MOCON, Modern Controls, Inc., Minneapolis, MN).

Demorest, R., Mayer, D.W., "Advanced measuring methods enhance WVTR testing," Packaging Technol. & Engineering, May/June 1996.

Elder, G., "Leak testing; A quality assurance NDT method," ASTM Standardization News, p. 46-49, Jan 1995.

European Pharmacopoeia (E.P.) VI.2.3.1. Rubber closures. Self-sealing test.

Frieben, W.R., Folck, R.J., Devisser, A., "Integrity testing of vial closure systems used for parenteral products," J. Parenter. Drug Assoc., Vol. 36, p. 112-116, 1982.

Gilchrist, J.E., Rhea, U.S., Dickerson, R.S., Campbell, J.E., "Helium leak test for micron-sized holes in canned foods," J. Food Protection, Vol. 48, No. 10, p. 856-860, 1985.

Gilchrist, J.,E., Shah, D.B., Radle, D.C., Dickerson, Jr., R.W., "Leak detection in flexible retort pouches," J. Food Protection, Vol. 52, No. 6, p. 412-415, Jun 1989.

Gnasasekharan, V., Floros, J.D., "A theoretical perspective on the minimum leak size," Chapter 8, Plastic Package Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

Gnasasekharan, V., Floros, J.D., " Automated nondestructive package integrity testing," Food Processing Automation II, Proceedings of the May 4-6,1992 conference, Lexington, KY, ASPE pub., 02-92, St. Joseph, MI, copy 1992.

Gnasasekharan, V., Floros, J.D., "Package integrity evaluation: Criteria for selecting a method," Packaging Technol. & Engineering, p. 44-48, Aug/Sep 1994.

Guazzo, D.M., "Package integrity testing," Chapter 4 of Parenteral Quality Control: Sterility, Pyrogen, Particulate, and Package Integrity Testing, 2nd ed., Akers, M.J., ed., Marcel Dekker, NY, copy 1994.

Guazzo, D.M., "Current approaches in leak testing pharmaceutical packages," PDA J. Pharm. Sci. & Technol., Vol., 50, No. 6, p. 378-385, Nov/Dec 1996.

Hansen, J., Jones, L., Anderson, H., Larsen, C., Miller, H., Scholla, M., Spitzley, J., Baldwin, A., "In quest of sterile packaging: Part I. Approaches to package testing," Medical Device & Diagnos. Ind., p. 56-61, Aug 1995.

Harper, C., "A microbial challenge procedure for identification of defective flexible plastic pouch seals," Chapter 10, Plastic Package Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

Health Industry Manufacturers Assoc. (HIMA), Report 78-411, "Microbial methods for assessment of package integrity," Washington, D.C., 1979.

Henke, C., Riech, R., "The current status of microbial-barrier testing of medical device packaging," Medical Device & Diagnos. Ind., p. 46-49, 94, Aug. 1992.

Hoffmann, J., "Computerized dry-air leak testing for process control," Medical Device & Diagnos. Ind., p. 204-208, Jan 1996.

ISO/DIS 11607, "Packaging for terminally sterilized medical devices," International Organization for Standardization, Geneva, Switzerland, 1995.

ISO 8362-2, "Injection containers for injectables and accessories - Part 2: Closures for injection vials," International Organization for Standardization, 1988.

Jones, L., Hansen, J., Anderson, H., Larsen, C., Miller, H., Scholla, M., Spitzley, J., Baldwin, A., "In quest of sterile packaging: Part 2. Physical package integrity testing methods," Medical Device & Diagnos. Ind., p. 81-85, Sep 1995.

Kirsch, L., Nguyen, L., Moeckly, C., "Pharmaceutical container/closure integrity I: Mass spectrometry-based helium leak rate detection for rubber-stoppered glass vials," J. Parenter. Sci. & Technol., Vol. 51, No. 5, p. 187-194, Sep./Oct. 1997.

Kirsch, L., Nguyen, L., Moeckly, C., Gerth, R., "Pharmaceutical container/closure integrity II: The relationship between microbial ingress and helium leak rates in rubber-stoppered glass vials," J. Parenter. Sci. & Technol., Vol. 51, No. 5, p. 195-202, Sep./Oct. 1997.

Kirsch, L., Nguyen, L., Gerth, R., "Pharmaceutical container/closure integrity III: Validation of the helium leak rate method for rigid pharmaceutical containers," J. Parenter. Sci. & Technol., Vol. 51, No. 5, p. 203-207, Sep./Oct. 1997.

Korczynski, M., "Evaluation of closure integrity," Aseptic Pharmaceutical Manufacturing: Technology for the 1990's, W.P. Olson, ed., Interpharm Press, Buffalo Grove, IL, copy 1987.

Leshniowski, W., Novak, R., Morrissey, R., "Assessment of microbial barrier properties of packaging materials," Chapter 30, Developments in Industrial Microbiology, Vol. 22, p. 381-392, 1980.

Lin, R.C., King, P.H., Johnson, M.R., "Examination of closures for integrity," Chapter 22, FDA Bacteriological Analytical Manual, 7th ed., p. 289-367, 1982.

Ludwig, J.D., Nolan, P.D., Davis, C.W., "Automated method for determining Instron residual seal force of glass vial/rubber closure systems," J. Parenter. Sci & Technol., Vol. 47, No. 5, p. 211-253, Sep/Oct 1993.

Ludwig, J.D., Davis, D.W., "Automated method for determining Instron residual seal force of glass vial/rubber closure systems Part II. 13 mm vials," PDA J. of Pharm. Sci. & Technol., Vol. 49, No. 5, p. 253-256, Sep/Oct 1995.

Marcy, J., "Integrity testing and biotest procedures for heat sealed containers," Chapter 6 of Plastic Packaging Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

McEldowney, S., Fletcher, M., "The effect of physical and microbiological factors on food container leakage," J. Applied Bacteriol., Vol. 69, p. 190-205, 1990.

McVean, D.E., Tuerck, P.A., Christenson, G.L., Cartensen, J.T., "Inadequacies in leakage test procedures for flame-sealed ampules," J. Pharm. Sci., Vol. 61, p. 1609-1611, 1972.

Moldenhauer, J., Bass, S., Kupinski, M., Walters, M., Rubio, S., "Microbial barrier assessment of tyvek stopper packaging for rubber closures," J. Pharm. Sci. & Technol., Vol. 50, No. 6, p. 391-398, Nov./Dec. 1996.

Morton, D.K., "Container/closure integrity of parenteral vials," J. Parenter. Sci. & Technol., Vol. 41, No. 5, p. 145-158, Sep/Oct 1987.

Morton, D.K., Lordi, N.G., "Residual seal force measurement of parenteral vials. I. Methodology," J. Parenter. Sci. & Technol., Vol. 42, No. 1, p. 23-29, Jan/Feb 1988.

Morton, D.K., Lordi, N.G., "Residual seal force measurement of parenteral vials. II. Elastomer Evaluation," J. Parenter. Sci. & Technol., Vol. 42, No. 2, p. 57-61, Mar/Apr 1988.

Morton, D.K., Lordi, N.G., Ambrosio, T.J., "Quantitative and mechanistic measurements of container/closure integrity. Leakage quantitation," J. Parenter. Sci. & Technol., Vol. 43, No. 2, p. 88-97, Mar/Apr 1989.

Morton, D.K., Lordi, N.G., Troutman, L.H., Ambrosio, T.J., "Quantitative and mechanistic measurements of container/closure integrity. Bubble, liquid and microbial leakage tests," J. Parenter. Sci. & Technol., Vol. 43, No. 3, p. 104-108 May/Jun 1989.

PDA, "Aspects of container/closure integrity," PDA Technical Information Bulletin No. 4, PDA, Bethesda, MD, pub, copy 1983.

Placencia, A.M., Peeler, J.T., "Package integrity of medical devices using microbial test methods and comparative studies using physical methods," World Conference on Packaging 89., Hamburg, Proceedings of the IAPRI Congress, Sec. 15.2, p. 531-540, Sep 27-29, 1989.

Placencia, A.M., Arin, M.L., Peeler, J.T., Oxborrow, G.S., "Physical tests are not enough," Medical Device & Diagnos. Ind., Sep 1988.

Reich, R., "Packaging films: A method of microbial barrier evaluation," Medical Device & Diagnos. Ind. p. 19-21, Feb 1986.

Reich, R., R., "A method for evaluating the microbial barrier properties of intact packages," Medical Device & Diagnos. Ind., p. 80-88, Mar 1985.

Rhea, U.S., Gilchrist, J.E., Peeler, J.T., Shah, D.B., "Comparison of helium leak test and vacuum leak test using canned foods: Collaborative study," J. Assoc. Off. Analytical Chemists, Vol. 67, No. 5, p. 942-945, 1984.

Rice, J., "Nondestructive testing of flexible lid seal integrity," Food Processing, p. 70-71, Sep 1986.

Rose, D., "Microperfusion permeation testing," Chapter 11, Plastic Package Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

Shaw, E.E., Litchfield, J.B., Reid, J.F., Harper, C.L., "Detection of defects in plastic laminated foil pouch packages: Machine vision sensing," Chapter 15, Plastic Package Integrity Testing. Assuring Seal Quality, B. Blakistone, C. Harper, ed., Institute of Packaging Professionals/Food Processor's Institute, pub., copy 1995.

Snow, D.W., "Microprocessor based systems for leak and flow testing," Medical Device & Diagnos. Ind., p. 104-107, Apr 1992.

Spitz, J., "The relevance of microbial challenge testing to package sterility," Medical Device & Diagnos. Ind., p. 142-148, Jan 1994.

Spitzley, J., "How effective is microbial challenge testing for intact sterile packaging?" Medical Device & Diagnos. Ind., p. 44-46, Aug 1993.

Stauffer, T., "Non-destructive in-line detection of leaks in food and beverage packages—An analysis of methods," J. Packaging Technology, Vol. 2, No. 4, p. 147-149, Aug 1988.

Tallentire, A., Sinclair, C.S., "Microbiological barrier properties of uncoated and coated spunbonded polyolefin (tyvek)," Medical Device & Diagnos. Ind., p. 51-54, Oct 1985.

The Parenteral Society, "The prevention and detection of leaks in ampoules, vials and other parenteral containers," Technical Monograph No. 3, The Parenteral Society, pub, Wiltshire, Great Britain, copy 1992.

United States Pharmacopoeia (USP) 23, <71> Sterility Tests, United States Pharmacopeial Convention, Inc., Rockville, MD, copy 1994.

United States Pharmacopoeia (USP) 23, <661> Containers, United States Pharmacopeial Convention, Inc., Rockville, MD, copy 1994.

United States Pharmacopoeia (USP) 23, <671> Containers - Permeation, United States Pharmacopeial Convention, Inc., Rockville, MD, copy 1994.

United States Pharmacopoeia (USP) 23, <1211> Sterilization and Sterility Assurance of Compendial Articles, United States Pharmacopeial Convention, Inc., Rockville, MD, copy 1994.

United States Pharmacopoeia (USP) Pharmacopoeial Forum, <1207> Sterile Product Packaging - Integrity Evaluation, Vol. 23, No. 6, p. 5066-5068, Nov.-Dec. 1997.

Varian Vacuum Products, Introduction to helium mass spectrometer leak detection, Varian Vacuum Products, Lexington, MA, pub., copy 1995.

Villiger, T., "Leakage testing of ampules. Validation method for high voltage pinhole detectors," Pharm. Ind., Vol. 49, p. 1071-1074, 1987.

Wang, Y.J., Chen, H., Busch, M., Baldwin, P., "Headspace analysis for parenteral products: Oxygen permeation and integrity test," Pharm. Technol., Vol. 21, No. 3, p. 108-122, Mar 1997.

Wu, V., Crist, B., "An automated method for vacuum testing pharmaceutical vials," J. Parenter. Sci & Technol, Vol. 43, No. 4, p. 179-182, Jul/Aug 1989.

Yam, K., "On-line, nondestructive system impacts integrity of pouches," Packaging Technol. & Engineering, p. 46-49, Jun/Jul 1995.

PDA Journal of Pharmaceutical Science and Technology

Supplement, July–August 1998 Volume 52

EDITOR: Joseph B. Schwartz

1998 OFFICERS AND DIRECTORS

Chairman: Joyce H. Aydlett
Chairman-Elect: Robert B. Myers
Secretary: Floyd Benjamin
Treasurer: R. Michael Enzinger, Ph.D.
Immediate Past Chairman: Raymond Shaw, Jr., Ph.D.

James E. Akers, Ph.D.
Jennie Allewell
Joyce L. DeYoung, Ph.D.
Stephanie R. Gray
Frederick A. Gustafson
Martin W. Henley
Henry K. Kwan, Ph.D.
Nikki V. Mehringer
Robert F. Morrissey, Ph.D.
Terry E. Munson
Toshiaki Nishihata, Ph.D.
Glenn E. Wright

President: Edmund M. Fry

PDA Journal of Pharmaceutical Science & Technology (ISSN 1076-397X) is published bimonthly by the PDA, Inc., 7500 Old Georgetown Rd., Suite 620, Bethesda, MD 20814.

Subscriptions—Membership dues in the PDA, Inc., include an annual subscription to the Journal for each member and corporate representative. For an application form and information regarding membership, address the Association. Industrial, university and public libraries and government agencies may subscribe at the rate of $135.00 per year. Back issues of the journal are available from the Association for $55.00 plus $1.50 postage and handling per in stock reprint; or $65.00 plus $3.00 postage and handling for photocopies of reprints not in stock.

Claims—Issues lost in transit will not be replaced if claim is received more than 90 days from date of issue or if loss was due to failure to give notice of change of address. The Association cannot accept responsibility for delivery outside the United States when shipment has been made by first-class mail.

Periodicals postage paid at Bethesda, Maryland and additional mailing offices. Postmaster: send address changes to the PDA Journal of Pharmaceutical Science & Technology, 7500 Old Georgetown Rd., Suite 620, Bethesda, MD 20814.

Printed in the U.S.A.

Formerly the
"Journal of Parenteral Science and Technology"

Copyright—PDA, Inc. 1998
ISSN 1076-397X